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Project 3 Milestone 3 White Paper

Screen Time & Associated Health Risks in Youths

Business Problem

This project evaluates whether higher daily screen exposure in youth is associated with worse sleep quality, higher mental health burden, and weaker attention and cognitive outcomes. Families, schools, and public health teams need evidence-based thresholds for when screen behavior becomes high risk and which usage patterns should be prioritized for intervention.

Background/History

Youth screen use has expanded rapidly across smartphones, social platforms, streaming content, and educational applications. Prior research reports mixed effect sizes, but a consistent theme is that heavy and unstructured use can displace sleep and physical activity while increasing emotional strain in some groups.

Data Explanation (Data Prep/Data Dictionary/etc.)

The draft dataset harmonizes youth-level records using a consistent schema: exposure variables (total and subtype screen hours), confounders (age, sex, socioeconomic index, physical activity), and outcomes (sleep, anxiety, depression, attention, cognition). Data preparation includes missing-value review with median imputation for low-rate numeric

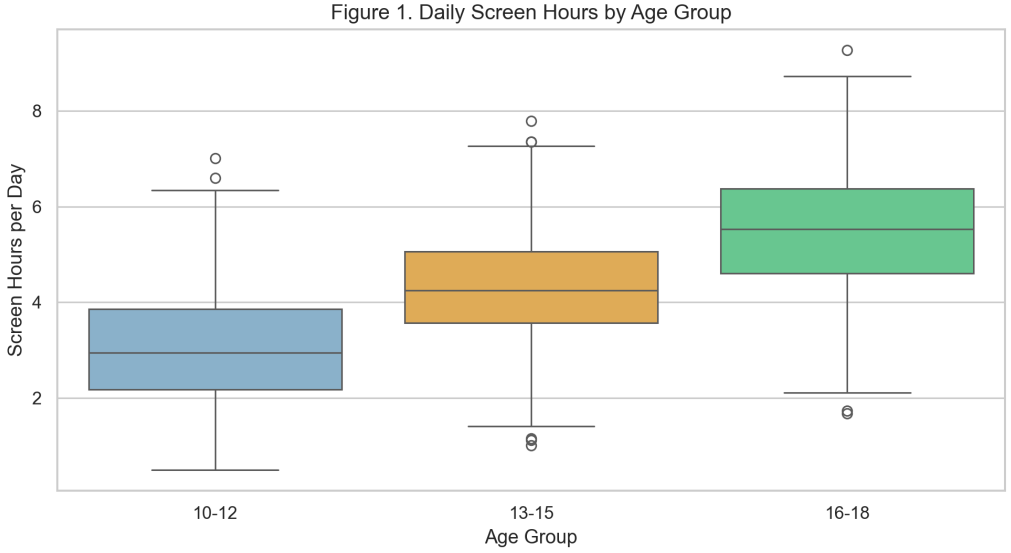
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gaps, outcome standardization for a mental-health index, and binary indicators (poor sleep and heavy screen exposure) for practical interpretation.

Methods & Analysis

We first profile distributions and subgroup differences by age band to identify where screen exposure concentrates. Figure 1 summarizes daily screen hours by age group and shows a clear shift toward higher exposure in older youth cohorts.

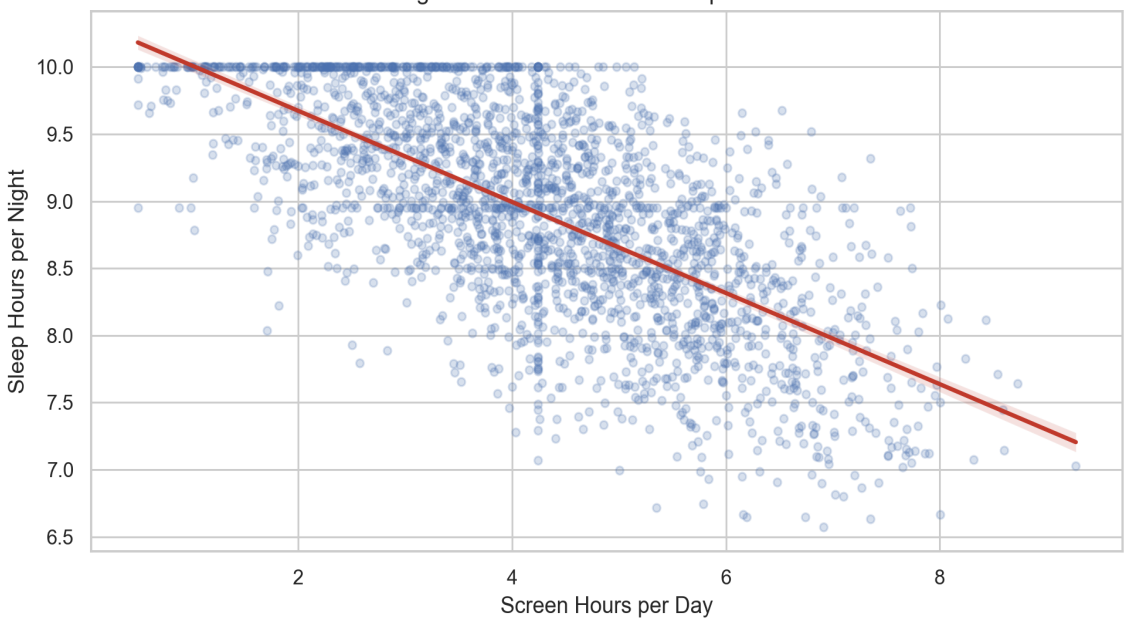
Figure 1. Daily Screen Hours by Age Group



Next, we quantify directional relationships with correlation and trend diagnostics. Figure 2 shows the negative slope between screen hours and sleep duration, and Figure 3

reports multivariable effect estimates from the adjusted model for mental-health burden.

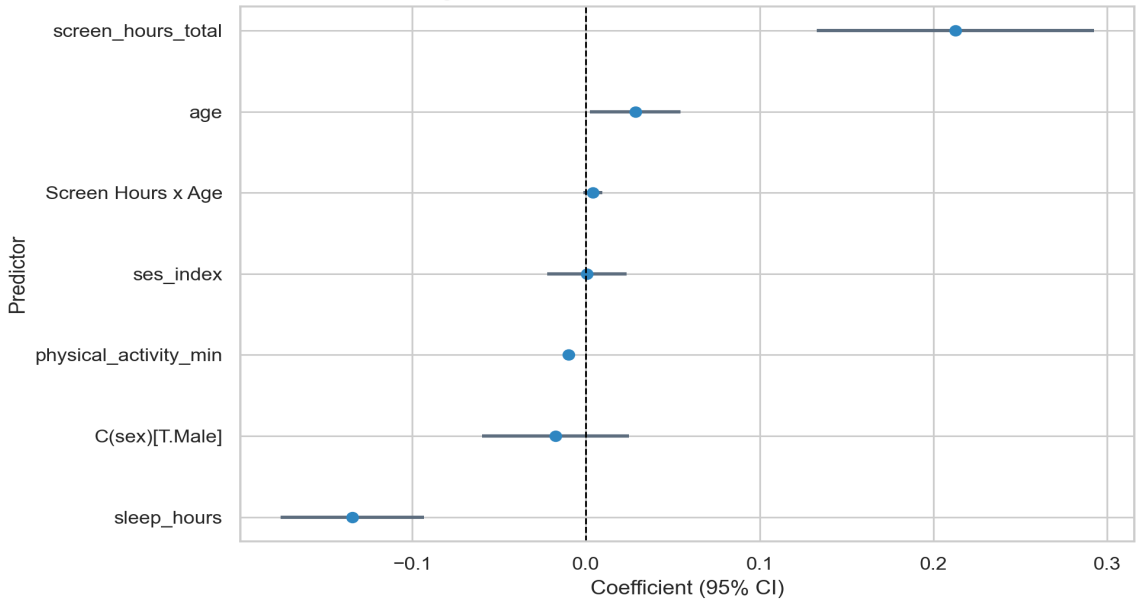
Figure 2. Screen Time vs Sleep Duration



For inference, we estimate OLS for continuous mental-health index and logistic regression for poor sleep risk while controlling for age, sex, SES, and activity. Draft results indicate higher screen exposure is associated with worse sleep and higher adjusted mental-health burden, with effect sizes large enough to inform targeted interventions.

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Figure 4. OLS Effect Estimates for Mental Health Index



Conclusion

The draft analysis supports the core hypothesis: heavier youth screen exposure is associated with poorer sleep and less favorable behavioral-health outcomes. These patterns remain directionally consistent after adjustment and provide a defensible basis for stakeholder guidance.

Because this is an observational draft, findings should be interpreted as associations rather than causation. Even so, the practical signal is strong enough to recommend prioritizing high-risk profiles (heavy use, low sleep, low activity) and deploying age-sensitive interventions in school and family settings.

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Assumptions: self-reported behaviors are sufficiently stable for group-level inference; harmonized variables capture key relationships. Limitations: cross-sectional structure, potential reporting bias, and residual confounding. Challenges: differing definitions of screen time across sources and non-uniform outcome scales.

Future Uses/Additional Applications: school risk dashboards, caregiver guidance tools, and intervention scenario testing. Recommendations and Implementation Plan: complete final Kaggle ingestion, rerun sensitivity checks, and publish tiered stakeholder guidance. Ethical Assessment: preserve de-identification, avoid stigmatizing narratives, and communicate uncertainty transparently. Appendix A (supporting documentation) is included in the PDF package.

Works Cited:

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